M13 Batch 10 — Full Set

M13.8 Instruments (ATA 31) · PDF pages 918–1008

GPWS modes and windshear, enhanced GPWS and terrain awareness, vibration monitoring and the fuel quantity indicating system.

208 questions · Level 3: 64 · Level 2: 103 · Level 1: 41

● marks a question built on a fact your instructor highlighted or underlined in the manual (152 of 208). Topics run in manual page order; inside each topic the questions are grouped by level, and the answers with explanations follow at the end of that topic.

No calculations, no unit conversions, no worked examples, nothing type-specific unless it was marked.

GPWS mode 6, mode 7 and windshear

PDF pages 919–927 · 34 questions

LEVEL 1 — RECALL

1. What makes the EGPWS give Mode 6 alerts and call-outs ●
   a. The aircraft climbing above predefined altitudes
   b. The aircraft descending below predefined altitudes
   c. The aircraft exceeding a predefined airspeed

2. Which indications do Mode 6 alerts and call-outs give to the crew ●
   a. Visual warnings and aural messages only
   b. Visual warnings and ARINC 429 outputs only
   c. Aural messages and ARINC 429 outputs only

3. Which flight phases does the Mode 7 windshear alerting cover ●
   a. Take-off and final approach
   b. Cruise and initial descent
   c. Taxi and initial climb

4. Between which heights above ground is the windshear detection active ●
   a. 50 and 2500 feet AGL
   b. 10 and 1500 feet AGL
   c. 200 and 1000 feet AGL

5. Airports that need envelope modulation are kept in the GPWC memory, so how are they identified there? ●
   a. By their runway heading alone
   b. By their ICAO four-letter code
   c. By their latitude and longitude

6. What does the terrain awareness alerting function use to predict a conflict ahead ●
   a. The aircraft position and altitude with an internal database
   b. The radio altimeter signal with the nearest runway length
   c. The weather radar returns with the aircraft ground speed

7. Terrain awareness is working normally, so which two systems supply its aeroplane data? ●
   a. The FMS and the WXR
   b. The ADIRS and the GPS
   c. The ADC and the ILS

8. Which alerting envelopes does the terrain alerting function compute ●
   a. A caution-level envelope and a warning-level envelope
   b. A take-off envelope and a final approach envelope
   c. A look-up envelope and a lateral offset envelope

LEVEL 2 — UNDERSTAND AND EXPLAIN

9. Where does the minimums call-out take its decision height from
   a. The setting on the EFIS control panels
   b. The runway elevation in the terrain database
   c. A fixed preset inside the radio altimeter

10. An approaching minimums call-out is selected, so how far above the decision height does it normally come on? ●
    a. 130 feet
    b. 100 feet
    c. 80 feet

11. One of the approaching call-outs is given with its own altitude selection instead of the normal trigger height, so which call-out is that? ●
    a. plus hundred
    b. approaching minimums
    c. approaching decision height

12. The windshear pre-alert comes up on take-off, so which indications does it give? ●
    a. ARINC 429 outputs and a stick shaker actuation
    b. Visual indications and ARINC 429 outputs
    c. Visual indications and a flashing master warning light

13. What does the measured windshear value represent
    a. The vector sum of inertial and air mass accelerations
    b. The rate of change of barometric altitude
    c. The difference between ground speed and airspeed

14. Which further aeroplane parameters does the GPWC check near an airport listed for envelope modulation ●
    a. Ground speed, roll attitude and calculated flight path angle
    b. Vertical speed, true heading and terrain elevation data
    c. Radio altitude, magnetic track and glide slope deviation

15. Which navigational signals can the envelope modulation feature use
    a. Localizer and glide slope beam deviation signals
    b. Updated VOR and DME range and bearing signals
    c. GPS or updated flight management system signals

16. Which sources does terrain awareness use for latitude and longitude ●
    a. GPS or FMS
    b. GPS or WXR
    c. ADIRS or ADC

17. Which of the terrain awareness input quantities is worked out by the EGPWC itself ●

- **a.** Ground track
- **b.** Flight path angle
- **c.** Barometric altitude

18. What does the EGPWC compare with its terrain database to find a threat ●

- **a.** The aeroplane position and track
- **b.** The aeroplane speed and flap setting
- **c.** The aeroplane pitch and roll

19. How does the terrain display show terrain height relative to the aeroplane ●

- **a.** By solid contour lines
- **b.** By numeric height labels
- **c.** By different coloured dots

20. The caution level aural has just been heard and the envelope is still penetrated, so when does it come again? ●

- **a.** Every three seconds
- **b.** Every seven seconds
- **c.** Without any pause

21. The aeroplane stays inside the terrain warning envelope, so how is PULL UP given? ●

- **a.** Twice and then once per 20% height loss
- **b.** Continuously
- **c.** Every seven seconds

22. What are the alert discretes of the look-ahead function provided for ●

- **a.** To reset the radio altimeter output
- **b.** To energise the terrain display relays
- **c.** To inhibit the weather radar transmitter

23. By which three dimensions are the terrain alerting envelopes sized

- **a.** A ground speed, a flight path angle and a roll attitude
- **b.** A runway distance, a radio altitude and a ground track angle
- **c.** A look-ahead distance, an altitude offset and a lateral distance

LEVEL 3 — APPLY AND REASON

24. At 2500 feet the operator rejects the aural TWENTY FIVE HUNDRED, so which other aural can be selected instead? ●

- **a.** The aural MINIMUMS MINIMUMS
- **b.** The aural RADIO ALTITUDE
- **c.** The aural TOO LOW TERRAIN

25. The aeroplane is at 300 feet radio altitude and the roll is increased steadily, so at which angles does the call-out occur? ●

- **a.** 10, 20 and 30 degrees
- **b.** 15, 30 and 45 degrees
- **c.** 35, 40 and 45 degrees

26. On a low go-around at 60 feet radio altitude the pilot rolls to 20 degrees, so what does Mode 6 do? ●

- **a.** It gives the aural BANK ANGLE, BANK ANGLE
- **b.** It gives no roll call-out at that height
- **c.** It gives the aural MINIMUMS, MINIMUMS

27. A windshear warning is triggered on take-off, so how do its indications differ from those of the pre-alert? ●

- **a.** A visual indication is added
- **b.** An ARINC 429 output is added
- **c.** An aural indication is added

28. On final approach at 900 feet AGL the head wind increases quickly and a severe up draft is met, so what does Mode 7 do? ●

- **a.** It stays quiet, because the gear is not yet down
- **b.** It alerts only after the aeroplane is below 500 feet
- **c.** It alerts, because the height is inside the active band

29. The aeroplane is near a listed airport but the glide slope deviation and runway heading fail to match, so what does the GPWC conclude? ●

- **a.** A modulation condition exists anyway
- **b.** The terrain database must be in error
- **c.** No modulation condition exists

30. At some locations the top radio altitude limit of the mode 2 alert area is lowered, so what does that allow?

- **a.** To let the aeroplane descend faster without any alerts
- **b.** To let the aeroplane approach without lowering the gear
- **c.** To let the aeroplane fly closer to the terrain without alerts

31. An approach needs mode 5 annunciations higher than normal and with the gear still up, so which change makes that possible?

- **a.** The descent rate limit is increased and the alert area raised
- **b.** The alert area is increased and the gear down requirement removed
- **c.** The alert area is lowered and a gear down requirement added

32. Both GPS and FMS position data are lost in the cruise, so which EGPWS function can no longer work? ●

- **a.** The look-ahead comparison with the terrain database
- **b.** The barometric altitude correction in the ADC
- **c.** The radio altimeter height measurement function

33. An aural is heard and exactly seven seconds later the same words come again, so which envelope is still penetrated? ●

- **a.** The clearance floor envelope
- **b.** The caution envelope
- **c.** The warning envelope

34. How does the look-ahead distance change as the aeroplane accelerates, and for what reason

- **a.** It increases, to keep the alerting time about equal
- **b.** It decreases, to keep the envelope narrow
- **c.** It stays fixed, because altitude alone sets it

Answers — GPWS mode 6, mode 7 and windshear

1. **b** Mode 6 is triggered by the aircraft descending below predefined altitudes such as DH and minimums. (L1, PDF p. 919)

2. **c** Mode 6 produces aural messages and ARINC 429 outputs but no visual indication. (L1, PDF p. 919)
3. **a** Mode 7 alerts during take-off and final approach. (L1, PDF p. 921)
4. **b** Detection is active between 10 and 1500 feet AGL. (L1, PDF p. 921)
5. **c** Those airports are held in memory identified by latitude and longitude. (L1, PDF p. 923)
6. **a** Geographic position and altitude are compared with the internal terrain database. (L1, PDF p. 925)
7. **b** Aeroplane data comes from the ADIRS and the GPS. (L1, PDF p. 925)
8. **a** Two envelopes are computed, one for caution level and one for warning level. (L1, PDF p. 927)
9. **a** The minimums call-out uses the decision height set on the EFIS control panels. (L2, PDF p. 919)
10. **c** The approaching minimums call-out normally comes on 80 feet above the decision height. (L2, PDF p. 919)
11. **a** Of the three approaching call-outs, plus hundred is the one with its own altitude selection, set for decision height + 100 feet, instead of the normal 80 feet above the decision height. (L2, PDF p. 919)
12. **b** The windshear caution, or pre-alert, provides visual and ARINC 429 output indications. (L2, PDF p. 921)
13. **a** The value is the vector sum of inertial versus air mass accelerations along and across the flight path. (L2, PDF p. 921)
14. **c** Radio altitude, magnetic track and glide slope deviation are all among those inputs. (L2, PDF p. 923)
15. **c** The feature uses GPS or updated FMS navigational signals. (L2, PDF p. 923)
16. **a** Latitude and longitude come from GPS or FMS. (L2, PDF p. 925)
17. **b** Only the flight path angle is noted as calculated by the EGPWC; the others are received. (L2, PDF p. 925)
18. **a** Position and track are compared with the database. (L2, PDF p. 925)
19. **c** Terrain height relative to the aeroplane is shown by different coloured dots. (L2, PDF p. 925)
20. **b** The caution message repeats every seven seconds while the envelope is penetrated. (L2, PDF p. 927)
21. **b** PULL UP repeats continuously for as long as the warning envelope is penetrated. (L2, PDF p. 927)
22. **b** The discretes energise the terrain display relays. (L2, PDF p. 927)
23. **c** The three sizing dimensions are look-ahead distance, altitude offset and lateral distance. (L2, PDF p. 927)
24. **b** At 2500 feet the alternative to TWENTY FIVE HUNDRED is the aural RADIO ALTITUDE. (L3, PDF p. 919)
25. **c** Above 130 feet the bank angle call-out comes at 35, 40 and 45 degrees. (L3, PDF p. 919)
26. **a** 60 feet lies between 30 and 130 feet and 20 degrees is larger than 10 degrees, so the call-out is given. (L3, PDF p. 919)
27. **c** The warning adds the aural to the visual and ARINC 429 indications the caution already gives. (L3, PDF p. 921)
28. **c** 900 feet lies inside the 10 to 1500 feet AGL detection band, so Mode 7 gives its alert. (L3, PDF p. 921)
29. **c** Position alone is not enough; the extra parameters must also confirm the condition. (L3, PDF p. 923)
30. **c** Lowering the mode 2 area lets the aeroplane fly closer to the terrain without alerts. (L3, PDF p. 924)
31. **b** Mode 5 modulation increases the alert area limits and removes the gear down requirement. (L3, PDF p. 924)
32. **a** Without latitude and longitude there is no position to compare with the terrain database. (L3, PDF p. 925)
33. **b** A seven second repeat is the caution level; the warning level repeats continuously. (L3, PDF p. 927)
34. **a** The distance grows with ground speed so that alerting times stay roughly the same. (L3, PDF p. 927)

Enhanced GPWS and terrain awareness

PDF pages 929–939 · 25 questions

LEVEL 1 — RECALL

35. Which colour paints ground that violates the terrain caution envelope ●

a. 50 per cent magenta
b. 100 per cent red
c. 100 per cent yellow

36. Which colour paints ground that violates the terrain warning envelope ●

a. 100 per cent yellow
b. 50 per cent green
c. 100 per cent red

37. What does the TCF alert function build around the intended runway ●

a. An increasing terrain clearance envelope
b. A fixed glide slope deviation envelope
c. A circular windshear detection envelope

38. Which two sources feed the navigational displays through the select relays ●

a. The ADIRS and the flight management system
b. The EGPWC and the WXR transceiver
c. The GPWC and the radio altimeter

LEVEL 2 — UNDERSTAND AND EXPLAIN

39. Under which condition is terrain kept off the display

a. When it lies more than 2000 ft above the runway
b. When it lies more than 2000 ft below reference altitude
c. When it lies more than 1000 ft above reference altitude

40. On which three quantities are the terrain clearance floor alerts based ●

a. Aircraft heading, runway length and barometric altitude data
b. Aircraft location, nearest runway centre point and radio altitude
c. Ground speed, flight path angle and nearest terrain elevation

41. When is the terrain clearance floor active during take-off

a. When mode 4 protection is not available
b. When the flaps have been retracted
c. When the landing gear is still down

42. The aeroplane moves further away from the airport, so how does the height of the clearance envelope change? ●

a. It increases as the distance increases
b. It stays the same at every distance
c. It decreases as the distance increases

43. Which aeroplane data does the EGPWC compare with the clearance envelope data ●

a. Latitude, longitude and barometric altitude
b. Latitude, longitude and radio altitude
c. Heading, ground speed and radio altitude

44. On which data bus standard is the terrain and weather display data carried ●

- **a.** ARINC 429
- **b.** ARINC 453
- **c.** ARINC 629

45. Which equipment receives the data from the terrain weather select relays ●

- **a.** The display electronics units
- **b.** The flight warning computers
- **c.** The air data computers

46. What happens when the TERR switch is selected a second time

- **a.** The relay deenergises and the weather radar is connected
- **b.** The relay deenergises and a test pattern is shown
- **c.** The relay stays energised and the terrain stays shown

LEVEL 3 — APPLY AND REASON

47. The aeroplane is in level flight and an area ahead is painted high density red, so where does that ground lie?

- **a.** More than 2000 ft below the aeroplane
- **b.** Between 1000 and 2000 ft above it
- **c.** More than 2000 ft above the aeroplane

48. A QFE approach is flown into a field with an extreme surface temperature, so why does the EGPWS need no special procedure?

- **a.** The radio altimeter replaces the air data input
- **b.** Corrected barometric altitude is used instead
- **c.** Its computed altitude copes with both conditions

49. Why does the terrain awareness function still work properly when the altimeter subscale has been set wrongly

- **a.** The air data computer recomputes the corrected value
- **b.** The radio altimeter replaces every altitude input
- **c.** Its computed altitude is not susceptible to that error

50. The aeroplane flies through a strong inversion layer, so why can the corrected barometric altitude mislead the EGPWS?

- **a.** It freezes at the last valid value
- **b.** It can be in error against the true height
- **c.** It is replaced by uncorrected altitude

51. The landing gear is lowered in the descent, so how does the lower edge of the medium density yellow band change?

- **a.** From 500 ft below to 250 ft below the reference altitude
- **b.** From 1000 ft below to 500 ft below the reference altitude
- **c.** From 250 ft below to 500 ft below the reference altitude

52. The landing gear is up and ground lies 400 ft below the reference altitude, so in which band is it painted?

- **a.** Medium density yellow
- **b.** Light density green
- **c.** Medium density green

53. TERR is selected on an EFIS control panel, so what does the select relay do?

- **a.** It energises and connects the hazard bus to the display units
- **b.** It deenergises and connects weather data to the display units
- **c.** It energises and connects terrain data to the display units

54. What do the transfer relays do while the weather radar is on and the caution criteria become fulfilled

- **a.** They connect both buses to the display units at once
- **b.** They drop the terrain bus and switch to the weather bus
- **c.** They drop the weather bus and switch to the terrain bus

55. A predictive windshear alert arrives while the terrain picture is up, so which picture is then shown?

- **a.** The weather one, with the shear data on it
- **b.** The terrain test pattern
- **c.** The terrain one, with the shear data on it

56. The terrain display relays have been energised by the alert discretes, so what do the navigational displays then carry? ●

- **a.** The terrain test pattern in cyan
- **b.** The conflicting ground in yellow or red
- **c.** The weather radar picture as before

57. A siren is followed by the same word three times during the climb-out at 300 feet, so which mode has alerted?

- **a.** Mode 2
- **b.** Mode 7
- **c.** Mode 5

58. The short level 1 test aural sequence begins with glideslope INOP, so what does that tell the engineer?

- **a.** The test was started below 2000 feet
- **b.** The glide slope lamp itself has failed
- **c.** No valid glide slope input is present

59. A level one test is run from the front panel of the computer, so why is the GPW module used instead for an operational test?

- **a.** The front panel test only checks the power supply
- **b.** The flight deck annunciations cannot be seen there
- **c.** The front panel test gives out no aural messages at all

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Answers — Enhanced GPWS and terrain awareness

35. **c** Caution envelope violations are painted solid yellow. (L1, PDF p. 929)

36. **c** Warning envelope violations are painted solid red. (L1, PDF p. 929)

37. **a** TCF builds a terrain clearance envelope that grows with distance from the runway. (L1, PDF p. 933)

38. **b** The two sources are the EGPWC and the weather radar transceiver. (L1, PDF p. 935)

39. **b** Terrain more than 2000 ft below reference altitude is not painted. (L2, PDF p. 929)

40. **b** Location, nearest runway centre point and radio altitude are the three inputs. (L2, PDF p. 933)

41. **a** In the take-off mode TCF covers the gap where mode 4 protection is missing. (L2, PDF p. 933)

42. **a** The envelope gets higher the further the aeroplane is from the airport. (L2, PDF p. 933)

43. **b** Latitude, longitude and radio altitude are compared with the envelope data. (L2, PDF p. 933)

44. **b** The display data runs on ARINC 453 buses. (L2, PDF p. 935)

45. **a** The relays feed the display electronics units, which drive the NDs. (L2, PDF p. 935)

46. **a** A second selection deenergises the relay and reconnects the weather radar. (L2, PDF p. 935)

47. **c** In level flight the reference altitude is the aircraft altitude, so high density red is more than 2000 ft above it. (L3, PDF p. 929)

48. **c** The geometric altitude function covers temperature extremes and QFE operation together. (L3, PDF p. 931)

49. **c** Geometric altitude is immune to an incorrect altimeter setting, so terrain awareness is unaffected. (L3, PDF p. 931)

50. **b** Under non-standard conditions such as inversions the corrected barometric altitude differs from the true altitude. (L3, PDF p. 931)

51. **a** With the gear up the band reaches 500 ft below, with the gear down only 250 ft below. (L3, PDF p. 932)

52. **a** With the gear up the medium density yellow band runs down to 500 ft below, so 400 ft below falls in it. (L3, PDF p. 932)

53. **c** Selecting TERR energises the relay and connects terrain data to the display units. (L3, PDF p. 935)

54. **c** With the criteria fulfilled the relays swap the weather bus out for the terrain bus. (L3, PDF p. 935)

55. **a** A shear alert overrides the terrain picture and reverts to the weather display. (L3, PDF p. 935)

56. **b** Once the relays are energised the conflicting ground is painted solid yellow or solid red. (L3, PDF p. 935)

57. **b** Only the mode 7 warning begins with a siren and repeats its word three times. (L3, PDF p. 938)

58. **c** The INOP annunciation in the sequence means the glide slope input is missing. (L3, PDF p. 939)

59. **b** From the computer front panel the flight deck annunciations are out of sight. (L3, PDF p. 939)

GPWS testing

PDF pages 941–945 · 7 questions

LEVEL 1 — RECALL

60. An engineer looks at the front of the ground proximity warning computer, so how many status LEDs are fitted there? ●

- **a.** Two LEDs
- **b.** Five LEDs
- **c.** Three LEDs

LEVEL 2 — UNDERSTAND AND EXPLAIN

61. When do the status LEDs on the front panel of the GPWC turn on

- **a.** When the aeroplane has left the ground
- **b.** When a self-test has been started from the cockpit
- **c.** When there is power to the computer

62. Which colour marks the external fault indication on the computer front panel ●

- **a.** Green
- **b.** Red
- **c.** Yellow

63. The green LED on the computer front panel is on, so what does that say? ●

- **a.** The computer is serviceable
- **b.** The computer has failed
- **c.** An external fault is present

64. Which colour marks a failure of the computer itself ●

- **a.** Red
- **b.** Yellow
- **c.** Green

LEVEL 3 — APPLY AND REASON

65. Level 2 reports No Faults but the engineer must still see what went wrong on earlier flights, so which level is needed?

- **a.** Level 6
- **b.** Level 4
- **c.** Level 3

66. The yellow LED is lit on the front panel while the green one stays on, so where should the engineer look first? ●

- **a.** Inside the computer itself
- **b.** At the equipment feeding the computer
- **c.** At the power supply to the computer

Answers — GPWS testing

60. **c** Three status LEDs are fitted on the front panel. (L1, PDF p. 945)

61. **c** The three status LEDs come on whenever the computer has power, not for any particular event. (L2, PDF p. 945)

62. **c** The external fault LED is yellow. (L2, PDF p. 945)

63. **a** Green is the computer OK indication. (L2, PDF p. 945)

64. **a** The computer fail LED is red. (L2, PDF p. 945)

65. **b** Current faults are level 2; the history of earlier legs is level 4. (L3, PDF p. 941)

66. **b** Yellow marks an external fault while green still says the computer is good, so the trouble is outside it. (L3, PDF p. 945)

Vibration measurement and indication

PDF pages 947–961 · 37 questions

LEVEL 1 — RECALL

67. What can be identified from the engine vibration indication ●

a. A rotor imbalance
b. An oil filter blockage
c. A fuel control fault

68. An engine vibration reading is seen in the cockpit, so how is its level expressed? ●

a. Usually in percent
b. Usually in units
c. Usually in millimetres

69. Which kind of transducer is used as an engine vibration sensor ●

a. A tachometer generator
b. An accelerometer
c. A strain gauge bridge

70. A technician counts the vibration sensors fitted to one engine, so how many does he normally find? ●

a. Four
b. Two
c. One

71. Which two types of accelerometer are used as engine vibration sensors ●

a. The electromagnetic type and the variable reluctance type
b. The piezoelectric-crystal type and the strain gauge type
c. The electromagnetic type and the piezoelectric-crystal type

72. What makes a piezoelectric crystal generate a voltage ●

a. A current passed through the crystal
b. A force applied to the crystal
c. A magnet moved across the crystal

73. A test switch is fitted to an analogue vibration display, so how much of it can be checked? ●

a. The filter setting only
b. The sensor wiring only
c. The entire indicator

74. A glass cockpit display shows the engine vibration, so how many vibration indications are given per engine? ●

a. Four
b. Two
c. One

75. An EICAS display shows the engine vibration, so how many values are given for each engine? ●

a. One
b. Two
c. Four

LEVEL 2 — UNDERSTAND AND EXPLAIN

76. Why does a rotor imbalance matter to the engineer ●

a. It always means a bearing has already failed
b. It can be the first sign of engine damage
c. It shows the engine is running overspeed

77. What do the engine vibration sensors deliver to the computer ●

a. Electrical signals
b. Hydraulic pressure signals
c. Pneumatic pressure signals

78. What does the computer do with the data it receives from the sensors ●

a. It turns it straight into a cockpit warning indication
b. It passes it unchanged to the flight data recorder
c. It filters it for indication and trend monitoring

79. Vibration sensors are not fitted to the engines alone on a helicopter, so which other parts carry them on many types? ●

a. Rotors and gearboxes
b. Flight control actuators
c. Fuel pumps and filters

80. Which quantity of the rotor system is sensed by an engine vibration pick-up ●

a. The radial acceleration of the rotor system
b. The axial displacement of the rotor system
c. The rotational speed of the rotor system

81. One of the two vibration sensors of an engine sits in the compressor area, so where is the other one? ●

a. In the turbine area
b. In the accessory gearbox area
c. In the combustion chamber area

82. How is the permanent magnet carried inside an electromagnetic vibration sensor ●

a. It is held in the centre by two springs, with a fixed coil around it
b. It is held on a base plate by an inertia mass, with a coil around it
c. It is held in the centre by a fixed coil, with two springs around it

83. The housing of an electromagnetic vibration sensor shakes with the engine, so what does the magnet inside do?

a. It turns with the rotor system
b. It stays almost still because of its inertia
c. It moves up and down with the housing

84. What produces the AC output voltage of an electromagnetic vibration sensor ●

a. The difference in motion between the coil and the magnetic field
b. The pressure of an inertia mass on the base plate
c. The force of the two springs acting on the permanent magnet

85. What is the job of the attuned filter inside an analogue vibration indicator

a. It amplifies the undesired frequencies
b. It screens out the undesired frequencies
c. It rectifies the critical frequencies

86. What does the marking REL AM on a vibration indicator tell the technician ●

a. That the scale carries no units and only follows a trend
b. That the scale shows the speed difference of the rotors
c. That the scale is graduated in mils of displacement

87. Which two rotor systems are covered by the two vibration indications of one engine ●

a. The N2 rotor system and the accessory drive system
b. The N1 rotor system and the fan rotor system
c. The N1 rotor system and the N2 rotor system

88. From how many sensor signals does the monitoring unit make the two vibration indications ●

a. From one single sensor signal on its own
b. From one sensor signal for each indication
c. From the two sensor signals added together

89. An engine vibration value is shown on the glass cockpit display, so which of the two engine vibration sensors normally feeds it? ●

a. The forward one
b. The aft one
c. Whichever reads higher

90. The forward vibration sensor of an engine fails, so what does the monitoring unit do? ●

a. It freezes the last indicated value
b. It removes the N2 indication only
c. It switches over to the aft one

91. In which two formats is the vibration value put on an EICAS display ●

a. In a digital and a bar graph format
b. In an analog and a colour coded format
c. In an analog and a digital format

92. A technician studies the vibration display, so what is put on it beside the value itself?

a. The stored limit values for the engine
b. The rotor speeds used for the comparison
c. The sensor and filter functions in use

LEVEL 3 — APPLY AND REASON

93. The vibration sensor near the N1 rotor shaft of an engine has failed, so which part of the engine has lost its pick-up? ●

a. The accessory gearbox area
b. The compressor area
c. The turbine area

94. A vibration signal reaches the monitoring unit with a high voltage but a low frequency, so what has it been told?

a. A small acceleration at a fast vibration
b. A large acceleration at a slow vibration
c. A large rotor speed at a slow acceleration

95. A technician wants the indication of a low frequency vibration on an older system, so which switch does he use?

- **a.** The one that picks the engine
- **b.** The one that picks the sensor
- **c.** The one that picks the filter

96. An older vibration system must show the oscillations of the add-on components, so which filter position is used?

- **a.** The low frequency one
- **b.** The broad band one
- **c.** The high frequency one

97. An analogue vibration indicator is driven by a moving coil, so what must be done to the alternating sensor signal before it can drive that movement?

- **a.** It must be attenuated
- **b.** It must be rectified
- **c.** It must be digitised

98. Why does an analogue vibration indicator need a filter selection at all

- **a.** Because motor and add-on vibrations fall in different frequency ranges
- **b.** Because the sensor output is an alternating current, not a direct one
- **c.** Because the scale carries no units of measurement, only a trend

99. The pointer of a vibration indicator sits below the zero mark, so what has happened?

- **a.** The filter has been switched off
- **b.** A system error has occurred
- **c.** The vibration has fallen to nothing

100. Why does the vibration monitoring system need the tachometer signals as well as the accelerometer signals

- **a.** Because the charge amplifiers must be supplied with power
- **b.** Because the tracking filter must follow the rotor speed
- **c.** Because the indicator scale must be set to the right range

101. How can one sensor signal be turned into separate N1 and N2 vibration indications ●

- **a.** By comparing its frequency with the two rotor speed signals
- **b.** By switching the filter between the forward and the aft pick-up
- **c.** By comparing its amplitude with the two stored limit values

102. A rotor speed signal is not available to the monitoring unit, so which indication mode is used?

- **a.** BB
- **b.** FAN
- **c.** LPT

103. Which actions may a pilot take after a vibration indication begins to pulse

- **a.** Checking the oil pressure, closing the bleeds or re-lighting the engine
- **b.** Selecting the aft pick-up, resetting the filter or changing the mode
- **c.** Scanning other engine indications, reducing power or shutting down

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Answers — Vibration measurement and indication

67. **a** The vibration indication reveals a rotor imbalance. (L1, PDF p. 947)

68. **b** The cockpit indication is normally given in units. (L1, PDF p. 947)

69. **b** The vibration sensors on an engine are accelerometers. (L1, PDF p. 949)
70. **b** Two sensors per engine is the usual fit. (L1, PDF p. 949)
71. **c** Only these two accelerometer types are used on engines. (L1, PDF p. 951)
72. **b** Applying force to the crystal is what produces the voltage. (L1, PDF p. 951)
73. **c** The test switch checks the whole indicator. (L1, PDF p. 955)
74. **b** Two indications are given for each engine. (L1, PDF p. 959)
75. **a** Only one value per engine is shown on this display. (L1, PDF p. 961)
76. **b** An imbalance can be the earliest sign that the engine is being damaged. (L2, PDF p. 947)
77. **a** The sensors send electrical signals to the computer. (L2, PDF p. 947)
78. **c** The computer collects and filters the data for both indication and trend monitoring. (L2, PDF p. 947)
79. **a** On many helicopters the rotors and gearboxes also carry vibration sensors. (L2, PDF p. 947)
80. **a** The accelerometer senses acceleration in the radial direction. (L2, PDF p. 949)
81. **a** The second of the two sensors is in the turbine area. (L2, PDF p. 949)
82. **a** Two springs hold the magnet centrally and the coil is the fixed part around it. (L2, PDF p. 951)
83. **b** Inertia keeps the magnet nearly stationary while the housing and coil move. (L2, PDF p. 951)
84. **a** Relative motion between coil and field induces the voltage, as in a generator. (L2, PDF p. 951)
85. **b** The attuned filter removes the frequencies that are not wanted. (L2, PDF p. 955)
86. **a** REL AM means an unitless scale meant only for trend following. (L2, PDF p. 955)
87. **c** One indication is for N1 and the other for N2. (L2, PDF p. 959)
88. **a** Both indications come from a single sensor signal. (L2, PDF p. 959)
89. **a** The forward sensor is the one normally used. (L2, PDF p. 959)
90. **c** On failure of the forward sensor the unit changes over to the aft sensor. (L2, PDF p. 959)
91. **c** The display gives the value both in analog and in digital form. (L2, PDF p. 961)
92. **c** The sensor and filter functions being used are shown with the value. (L2, PDF p. 961)
93. **b** The sensor near the N1 rotor shaft is the one in the compressor area, so that area is the one left unmonitored. (L3, PDF p. 949)
94. **b** Voltage carries the acceleration level and frequency carries the vibration frequency, so high voltage with low frequency is a big acceleration vibrating slowly. (L3, PDF p. 951)
95. **c** Frequency band is set by the filter switch, not by the sensor switch. (L3, PDF p. 953)
96. **a** Add-on components give the low oscillations, so the low frequency filter is selected. (L3, PDF p. 955)
97. **b** A moving coil instrument needs a unidirectional current, so the alternating signal is rectified before display. (L3, PDF p. 955)
98. **a** High oscillations come from the motor and low ones from the add-on components, so the filter has to be set to the range being looked at. (L3, PDF p. 955)
99. **b** A pointer below zero on the OFF / Error mark means a system error, not a zero vibration. (L3, PDF p. 957)
100. **b** The filter has to track the rotor speed, and it is the tachometer signals that tell it that speed. (L3, PDF p. 957)
101. **a** The frequency of the vibration signal is compared with the two rotor speed signals to sort it into N1 and N2. (L3, PDF p. 959)
102. **a** Without a rotor speed signal the filters cannot track, so the unfiltered broad band mode is used. (L3, PDF p. 961)
103. **c** The pilot has to look for the reason, by scanning other engine indications, reducing power or shutting the engine down. (L3, PDF p. 961)

Rotor and transmission monitoring

PDF pages 963–967 · 21 questions

LEVEL 1 — RECALL

104. Which three components make up the main rotor rpm monitoring system ●

a. A magnetic pick up, an rpm indicator and a triple tachometer
b. A magnetic pick up, an rpm indicator and an rpm warning unit
c. A magnetic pick up, an rpm warning unit and a master caution light

105. A technician must find the rotor rpm magnetic pick up, so where is it installed? ●

a. Within the tail rotor gearbox
b. Within the rotor mast
c. Within the main gear box

106. Which three components make up the mast moment monitoring system ●

a. A strain gauge bridge, a magnetic pick up and indicators
b. A strain gauge bridge, a sensor transmitting unit and indicators
c. A sensor transmitting unit, an rpm warning unit and indicators

107. A strain gauge bridge is fitted for the bending moment monitoring, so where does it sit? ●

a. In the rotor mast
b. In the rotor head
c. In the main transmission

LEVEL 2 — UNDERSTAND AND EXPLAIN

108. Which kind of sensor measures the main rotor speed ●

a. A piezoelectric type
b. A magnetic type
c. A capacitive type

109. What is the voltage peak from the rotor speed pickup proportional to ●

a. The frequency of the interrupter
b. The temperature of the main gear box
c. The movement of the interrupter

110. A pilot reads the rotor rpm on the triple tachometer, so which part of the dial shows it? ●

a. The small pointer R on the inner scale
b. The large pointer N1 on the outer scale
c. The digital window at the foot of the dial

111. The rotor rpm is shown permanently in two places in the cockpit, so which two are they? ●

a. The triple tachometer and the caution and advisory display
b. The rpm warning unit and the central engine display unit
c. The triple tachometer and the central engine display unit

112. Through which unit do the speed pickup signals pass on their way to the indicators

a. The RPM warning unit
b. The sensor transmitting unit
c. The central engine display unit

113. The rotor rpm falls below the operating range, so what is given to the crew? ●

a. A removal of the rpm indication
b. A visual warning only
c. A visual and an audio warning

114. Where are the sampled transmission data indicated constantly

a. On the analog instruments or on the Caution and Advisory Display
b. On the triple tachometer or on the rpm warning unit
c. On the analog instruments or on the mast moment indicator

115. Which two transmission parameters have their own instruments showing them constantly ●

a. The oil temperature and the oil pressure
b. The oil temperature and the chip detection
c. The oil pressure and the vibration level

116. A sampled transmission value goes outside its operational range, so what happens? ●

a. The indication is frozen at the limit value
b. The parameter is taken off the display
c. A warning appears on the respective warning unit

117. What does the mast moment monitoring system measure ●

a. The bending moment between the rotor mast and the main transmission
b. The twisting moment between the engine and the main transmission
c. The vibration between the rotor mast and the rotor head

118. Through which unit does the measured mast moment reach the indicators ●

a. The sensor transmitting unit
b. The caution and advisory display
c. The rpm warning unit

119. The maximum mast moment is exceeded in flight, so what comes with the warning light?

a. An automatic power reduction
b. An entry in the fault memory
c. An aural indication

120. How is an exceeded mast moment shown on an electronic mast moment indicator ●

a. By a red marking
b. By a warning light
c. By a flashing scale

121. A helicopter is not fitted with an EFIS, so what must be done to the mast moment indicator before each engine start? ●

a. It must be zeroed by the pilot
b. It must be calibrated
c. It must be tested against its warning light

122. A helicopter does carry an EFIS, so how is that mast moment indicator check carried out? ●

a. It is done by the pilot before each engine start
b. It is done automatically by the system
c. It is done by maintenance at the daily check

LEVEL 3 — APPLY AND REASON

123. A tail rotor gearbox develops a malfunction, so how will the crew learn of it? ●

- **a.** By a caution light only
- **b.** By its own analog instrument
- **c.** By a pointer leaving the green arc

124. A helicopter without EFIS was started without the bending load indicator being calibrated, so what is now in doubt? ●

- **a.** The supply to the strain gauge bridge
- **b.** The accuracy of the mast moment indication
- **c.** The aural warning of the rotor rpm

Answers — Rotor and transmission monitoring

104. b These three are the components of the system. (L1, PDF p. 963)

105. c It sits inside the main gear box. (L1, PDF p. 963)

106. b These three are the components listed for the system. (L1, PDF p. 967)

107. a The bridge is installed in the rotor mast. (L1, PDF p. 967)

108. b The rotor speed pickup is a magnetic sensor. (L2, PDF p. 963)

109. a The peak is in proportion to the interrupter frequency. (L2, PDF p. 963)

110. a Rotor rpm is the small R pointer on the inner scale. (L2, PDF p. 963)

111. c Both the triple tachometer and the central engine display unit carry the rotor rpm. (L2, PDF p. 963)

112. a The signals go through the rotor RPM warning unit. (L2, PDF p. 963)

113. c Going outside the range either way sets off both visual and audio warning. (L2, PDF p. 963)

114. a Either the analog indicators or the CAD carry them constantly. (L2, PDF p. 965)

115. a Transmission oil temperature and pressure are the two indicated constantly. (L2, PDF p. 965)

116. c Out-of-range data produce a warning on the respective warning unit. (L2, PDF p. 965)

117. a It is the bending moment between mast and main transmission. (L2, PDF p. 967)

118. a The signal passes through the sensor transmitting unit. (L2, PDF p. 967)

119. c An aural indication accompanies the visual one. (L2, PDF p. 967)

120. a On the electronic indicator it is a red marking; the warning light belongs to the analog one. (L2, PDF p. 967)

121. b Without an EFIS the indicator has to be calibrated before each engine start. (L2, PDF p. 967)

122. b With an EFIS fitted the check is done automatically. (L2, PDF p. 967)

123. a Gearboxes other than the main transmission are only covered by caution lights. (L3, PDF p. 965)

124. b Without the pre-start calibration the indicated mast moment can no longer be trusted. (L3, PDF p. 967)

Fuel quantity indication

PDF pages 969–977 · 15 questions

LEVEL 1 — RECALL

125. Which two methods are basically used to measure the fuel quantity ●

a. A capacitor system and an ultrasonic system
b. A capacitor system and a float switch system
c. An ultrasonic system and a flow meter system

126. What decides the number of fuel quantity probes in a tank ●

a. The size and shape of the aircraft
b. The size and shape of that tank
c. The number of pumps in that tank

127. Which electrical component is a fuel quantity probe ●

a. A resistor
b. An inductor
c. A capacitor

128. What is the material dependent factor that affects a probe's capacitance called ●

a. The dielectric constant
b. The specific gravity
c. The capacitive reactance

LEVEL 2 — UNDERSTAND AND EXPLAIN

129. A large aeroplane carries an extra fuel quantity indication outside the cockpit, so who needs it?

a. The loadmaster in the hold
b. The flight crew on the walk round
c. The fuelling personnel

130. On which aeroplanes is the ultrasonic fuel measuring principle used ●

a. On all of them
b. On larger ones only
c. On smaller ones only

131. How are the fuel quantity probes installed in a tank ●

a. Vertically, from the bottom right up to the top
b. Horizontally, lying along the floor of the tank
c. Vertically, from the top down to about half depth

132. A technician lays out the probes taken from one tank, so what does he notice about them? ●

a. They are of different length
b. They are of different diameter
c. They are all of the same length

133. The amount of fuel in a tank changes, so what changes at the probe? ●

a. Its capacity
b. Its resistance
c. Its inductance

134. How are the two tubes of a fuel quantity probe arranged

- **a.** Coaxially and electrically isolated from each other
- **b.** Side by side and electrically bonded to each other
- **c.** Coaxially and electrically bonded to each other

135. What is meant by the capacitance of a fuel quantity probe ●

- **a.** Its opposition to the flow of current
- **b.** Its capability to store electrical energy
- **c.** Its ability to generate a voltage

136. On which two features of the plates does the capacitance of a probe depend ●

- **a.** Their area and the thickness of them
- **b.** Their area and the distance between them
- **c.** Their length and the thickness of them

LEVEL 3 — APPLY AND REASON

137. The fuel level rises past the middle of a probe, so what has happened between its two tubes? ●

- **a.** The gap between the tubes has become smaller
- **b.** All of the air has been replaced by fuel
- **c.** Part of the air has been replaced by fuel

138. Two probes in the same tank are both completely covered by fuel yet give different values, so what explains it? ●

- **a.** Their plates differ in spacing because they differ in fuel cover
- **b.** Their plates differ in area because they differ in fuel cover
- **c.** Their plates differ in area because they differ in length

139. Why does a capacitive probe give a different reading as the tank fills ●

- **a.** Because the area of its tubes changes
- **b.** Because the spacing between its tubes changes
- **c.** Because the material between its tubes changes

Answers — Fuel quantity indication

125. a Capacitors and the ultrasonic principle are the two methods given. (L1, PDF p. 969)

126. b Tank size and shape set how many probes are needed. (L1, PDF p. 975)

127. c Each probe is a capacitor. (L1, PDF p. 975)

128. a It is named the dielectric constant. (L1, PDF p. 977)

129. c The indication at the fuelling station is there for the fuelling personnel. (L2, PDF p. 969)

130. b The ultrasonic method is restricted to larger aeroplanes. (L2, PDF p. 969)

131. a They stand vertically and run the full depth of the tank. (L2, PDF p. 975)

132. a Because the tank depth varies the probes come in different lengths. (L2, PDF p. 975)

133. a The probe's capacity follows the amount of fuel. (L2, PDF p. 975)

134. a They sit one inside the other and are isolated, which is what makes them a capacitor. (L2, PDF p. 977)

135. b Capacitance is the ability to store electrical energy. (L2, PDF p. 977)

136. b Plate area A and plate separation d are the two given. (L2, PDF p. 977)

137. c The gap is filled to the same level as the tank, so a half covered probe has fuel in part of it and air in the rest. (L3, PDF p. 977)

138. c Probes in one tank are of different lengths, so their plate areas differ even when both are fully covered. (L3, PDF p. 977)

139. c Filling the tank swaps air for fuel between the tubes, and that material change alters the capacitance. (L3, PDF p. 977)

Capacitance probes and compensation

PDF pages 979–989 · 48 questions

LEVEL 1 — RECALL

140. Which property of air and fuel does the capacitance method of quantity measurement make use of ●

a. Their differing boiling points
b. Their differing dielectric constants
c. Their differing electrical resistances

141. A bridge transformer has a centre tap and two secondary windings, so what phase relationship do the windings have to the tap? ●

a. Opposite
b. Identical
c. Ninety degrees apart

142. How is the immersion depth measured in the fuel quantity computer system ●

a. By a float and an LVDT at the tank units and the compensator
b. By travel time of a pulse at the tank units and the compensator
c. By capacity changes at the tank units and the compensator

143. Which two quantities does the cadensicon sensor measure separately for the computer ●

a. The density and the dielectric constant
b. The temperature and the specific gravity
c. The volume and the dielectric constant

144. Which four items make up the fuel quantity indicating system that compensates fuel scatter ●

a. Tank units, a totaliser, a repeater and a compensator
b. Tank units, a computer, a densitometer and a water detector
c. Tank units, a computer, a cadensicon and a compensator

145. A technician looks for the compensator inside a fuel tank, so where in the tank is it installed? ●

a. Near the highest point
b. At about mid height
c. Near the lowest point

LEVEL 2 — UNDERSTAND AND EXPLAIN

146. Why is the amount of fuel carried indicated in units of weight ●

a. The tank capacity is certified in terms of weight
b. The energy available from the fuel depends on volume
c. The energy available from the fuel depends on weight

147. What makes a direct weight indication of the fuel necessary on a large commercial aircraft ●

a. Fuel density is constant so the weight can be read straight from the volume
b. Fuel can be half the take-off weight and the approved weights need it
c. Fuel is always loaded by volume and the tank gauges are marked in litres only

148. The same number of litres of fuel is loaded on a very cold day and on a very hot day, so what differs between the two loads?

a. The weight on board
b. The dielectric constant of the fuel
c. The capacity of the empty tank

149. The fuel level in a tank alters, so what does the multi-tubular surface condenser in it do? ●

a. It changes its excitation frequency
b. It changes its capacity
c. It changes its resistance

150. What is the total capacity of the tank condenser made up of ●

a. The empty tank capacity minus the change from fuel displacing air
b. The change from fuel displacing air on its own
c. The empty tank capacity plus the change from fuel displacing air

151. What happens to the capacitance of a tank sensor as the tank fills with fuel ●

a. It stays constant
b. It increases
c. It decreases

152. A tank sensor has changed its capacitance as the tank filled, so where is that change used? ●

a. In a chopper circuit that gives a pulse train output
b. In a resolver circuit that gives a shaft angle output
c. In a bridge circuit that gives an amplified signal

153. What does the amplified signal from the capacitance bridge act on ●

a. The shutoff valve in the refuelling gallery line
b. The pointer or the digital display
c. The excitation winding of the tank unit sensor

154. What condition exists when the tank unit capacitance and the reference condenser capacitance are equal ●

a. Maximum current runs through the instrument
b. A current runs that drives the pointer to full
c. No current runs through the instrument

155. What is true of the two points across which the instrument is connected when the bridge is balanced ●

a. They are at the supply potential
b. They are at opposite electrical potential
c. They are at the same electrical potential

156. What does the position of the potentiometer arm represent once the bridge has been rebalanced ●

a. The ratio of the two capacitance values
b. The sum of the two capacitance readings
c. The supply voltage applied to the bridge

157. How is the capacitive bridge of a fuel quantity indicator rebalanced in service ●

a. Automatically, by amplifying the difference signal
b. Manually, by the crew turning a trimming knob
c. Automatically, by switching in reference condensers

158. The amplifier of a capacitive fuel gauge puts out a voltage, so what does that voltage drive? ●

a. A heater that trims the reference condenser
b. A solenoid that trips the warning flag
c. A motor that turns the potentiometer

159. Fuel is being burned off in flight and the gauge is following it down, so what brings the rebalancing motor to rest at each new reading? ●

a. The potentiometer reaching its end stop
b. The difference signal falling to zero
c. The amplifier output reaching its limit

160. Why does the relationship between dielectric constant and density have to be corrected ●

a. It varies with the altitude the aircraft is flown at
b. It varies with the age of the tank units in the tank
c. It varies with the fuel consistency of each country

161. How is the dielectric constant obtained inside a cadensicon sensor ●

a. From the travel time of an ultrasonic pulse
b. From the capacity change of a condenser
c. From the depth of immersion of a float

162. How is the density obtained inside a cadensicon sensor ●

a. By a thermistor bridge sensing the fuel temperature
b. By a condenser whose capacity changes with the fuel
c. By a cardanically suspended float of fixed weight

163. The cadensicon float has sunk to a certain depth in the fuel, so how is that depth turned into a signal for the computer? ●

a. By an LVDT
b. By a strain gauge
c. By a synchro resolver

164. How is density found in systems that damp a mass instead of floating one ●

a. From the force needed to hold the damped mass still
b. From the time the damped mass takes to come to rest
c. From the ratio of oscillations added to those removed

165. What does the fuel quantity indicating computer work out from the probes it monitors

a. The density of fuel in the related tank
b. The volume of fuel in the related tank
c. The mass of fuel in the related tank

166. When does the cadensicon sensor take its measurements of K and D ●

a. As the engines are started
b. As the tank runs nearly empty
c. As fuel enters the tank

167. In what form does the cadensicon deliver its two outputs to the computer ●

a. One electrical and one pneumatic
b. Both as electrical signals
c. Both as mechanical movements

168. A cadensicon sensor measures two properties of the fuel, so which one of them is sensed mechanically? ●

a. The dielectric constant
b. The density
c. The temperature

169. What is the fuel scatter effect ●

a. The K to D relationship is not constant between fuel samples
b. The K to D relationship alters as the fuel temperature changes
c. The capacitance of a probe drifts as the probe gets older

170. The fuel in a tank warms up over a long flight, so what happens to the relationship between K and D? ●

a. It is pulled back onto the average law
b. It is not influenced by temperature
c. It changes in step with the fuel density

LEVEL 3 — APPLY AND REASON

171. Water has collected between the plates of a capacitance probe, so how does the probe capacitance compare with the same probe covered in fuel? ●

a. Slightly lower
b. The same
c. Much higher

172. Water has collected in the bottom of a tank, so how would the quantity indication be affected? ●

a. It would read below the fuel actually present
b. It would stay at the correct reading
c. It would read above the fuel actually present

173. Why is the capacity of a tank condenser compared with its capacity when the tank is empty ●

a. The difference corresponds to the quantity of fuel present
b. The difference corresponds to the temperature of the fuel
c. The difference corresponds to the dielectric constant of the fuel

174. What is gained by leaving the tank unit as the only variable in the bridge

a. A lower supply current to the bridge
b. A faster response of the indication
c. Greater precision of the indication

175. The aircraft is being refuelled and the level in a tank is rising, so what is happening at the master indicator?

a. Tank unit current falls and the indication moves downscale
b. Compensator current rises and the indication holds steady
c. Tank unit current rises and the indication moves upscale

176. Why does the indicator run downscale when the fuel quantity test switch is operated

a. The compensator current that normally drives it to full is missing
b. The supply to the rebalancing motor is reversed
c. The tank unit current that normally drives it to full is missing

177. The rebalancing motor of a capacitive gauge has just run to a stop, so what is true of the two bridge currents at that moment?

a. They are both at zero
b. They are equal
c. They are in opposite phase

178. The master indicator already shows the figure it has worked out, so how do the repeater and the fuel summation unit come to show the same?

a. The synchronising voltage is sent to them from the totaliser
b. The computed value is sent to them over the compensator line
c. The tank unit current is sent to them over the coaxial cable

179. How does the built-in test of the computerised system differ from the test started by the flight deck switch

a. One runs automatically and tests the displays, the other codes faults
b. One runs automatically and codes faults, the other tests the displays
c. One tests the displays, the other codes faults found on the ground only

180. A computerised fuel quantity indicator is showing an error code instead of a quantity, so which of the computer circuits produced it?

a. The one sending the signal out to the tank units
b. The one feeding the repeater and the summation unit
c. The one running the automatic system check

181. A float sinks further in one fuel sample than in another of the same temperature, so what does that tell the computer? ●

a. That the two samples differ in dielectric constant
b. That the two samples differ in volume
c. That the two samples differ in density

182. Refuelling has to stop before a tank overflows, so what is built into the fuel quantity computer system for this? ●

a. A totaliser that sums all of the master systems
b. A preselection of the wanted load with a shutoff
c. A trimming condenser at the cockpit indicator

183. Why does the aircraft attitude matter to the preselection and shutoff function ●

a. The dielectric constant of the fuel depends on aircraft position
b. The point at which a tank overflows depends on aircraft position
c. The density measured by the float depends on aircraft position

184. What does the computer derive from the cadensicon and compensator data it monitors

a. The density of the fuel in the tanks
b. The temperature of the fuel in the tanks
c. The volume of the fuel in the tanks

185. The probes alone can only tell the computer how much space the fuel fills, so what else must it have before a figure in kilograms can be shown?

a. The density of the fuel
b. The temperature of the fuel
c. The usable capacity of the tank

186. The outer tanks still hold fuel from the previous flight when fresh fuel is loaded, so how does the computer handle the mixture?

a. It applies the values of the fresh fuel to both quantities
b. It ignores the old fuel until that tank has been emptied
c. It calculates a mean value from the old and the new fuel

187. Why can fuel taken on at two different airports upset the quantity indication unless the computer allows for it

- **a.** The two loads can have different K to D values
- **b.** The two loads can have different fuel temperatures
- **c.** The two loads can have different tank unit capacitances

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Answers — Capacitance probes and compensation

140. b The capacitance method uses the differing dielectric constants of air and fuel. (L1, PDF p. 979)

141. a The two secondary windings are in opposite phase to the centre tap. (L1, PDF p. 981)

142. c Immersion depth is measured by capacity changes using the tank units and the compensator. (L1, PDF p. 987)

143. a The cadensicon measures density and dielectric constant separately. (L1, PDF p. 987)

144. c The system has tank units, a fuel quantity indicating computer, a cadensicon sensor and a compensator. (L1, PDF p. 989)

145. c The compensators are installed near to the lowest point of the tanks. (L1, PDF p. 989)

146. c The energy present in the form of fuel is dependent on weight, so weight is indicated. (L2, PDF p. 979)

147. b Fuel can be up to one half of take-off weight, so direct weight indication is needed for the approved take-off and landing weights. (L2, PDF p. 979)

148. a Volume and specific gravity change with temperature, so the same number of litres can differ by several tonnes. (L2, PDF p. 979)

149. b The condenser changes its capacity depending on the level of fuel in the tank. (L2, PDF p. 979)

150. c Total capacity is the empty tank capacity plus the capacity change due to the displacement of air between the plates. (L2, PDF p. 979)

151. b Fuel has a dielectric strength more than twice that of air, so sensor capacitance rises with fuel content. (L2, PDF p. 979)

152. c The change in capacitance is used in a bridge circuit to provide an amplified signal. (L2, PDF p. 979)

153. b The amplified signal rotates the pointer or actuates the digital display in the indicator. (L2, PDF p. 979)

154. c With equal capacitances the two currents compensate each other and nothing flows through the instrument. (L2, PDF p. 981)

155. c At balance the two points have the same electrical potential, so no current flows. (L2, PDF p. 981)

156. a The potentiometer arm position indicates the ratio of the two capacitance values. (L2, PDF p. 981)

157. a Fuel quantity capacitive bridges are balanced automatically by amplifying the difference signal. (L2, PDF p. 981)

158. c The amplifier output voltage drives a motor which turns the potentiometer. (L2, PDF p. 981)

159. b The motor stops when the difference signal is zero. (L2, PDF p. 981)

160. c The ratio of K to D varies in accordance with fuel consistency in individual countries. (L2, PDF p. 987)

161. b K is calculated from the capacity change of a condenser. (L2, PDF p. 987)

162. c D is found mechanically by a cardanically suspended float of predetermined weight. (L2, PDF p. 987)

163. a The depth of immersion is measured by an LVDT and sent to the computer. (L2, PDF p. 987)

164. c In those systems the ratio of oscillations added to oscillations removed determines density. (L2, PDF p. 987)

165. b The computer uses signal conditioning on the probe signals to calculate the volume of fuel in the related tank. (L2, PDF p. 989)

166. c The cadensicon measures K and D separately as fuel enters the tank. (L2, PDF p. 989)

167. b Outputs K and D are both electrical signals. (L2, PDF p. 989)

168. b K is measured electrically only while D is measured mechanically by a float. (L2, PDF p. 989)

169. a Fuel scatter is the phenomenon that the relationship of K to D between different fuel samples is not constant. (L2, PDF p. 989)

170. b A change in temperature will not influence this relationship. (L2, PDF p. 989)

171. c Water is given a far larger dielectric constant than fuel, and the capacitance rises with the dielectric constant of whatever fills the probe. (L3, PDF p. 979)

172. c The capacitance follows the dielectric constant of what fills the probe, and water's is far above that of fuel, so the gauge reads more fuel than is there. (L3, PDF p. 979)

173. a The capacity above the empty tank value is the part caused by fuel, and that difference corresponds to the fuel quantity. (L3, PDF p. 979)

174. c Having only the tank unit variable increases indicator precision. (L3, PDF p. 983)

175. c A rising level raises the tank unit capacitance and current, and the quantity indicator moves upscale. (L3, PDF p. 983)

176. c Operating the test switch removes the tank unit current that normally moves the indicator towards FULL, so it runs downscale. (L3, PDF p. 983)

177. b The motor runs until the two currents are equal, and that is where it stops. (L3, PDF p. 983)

178. b The value shown on the master indicator is transmitted in a further circuit via the compensator line to the repeater and the fuel summation unit. (L3, PDF p. 985)

179. b The built-in test runs automatically and shows any fault as an error code, while the FQ test switch starts a display test of all indicators. (L3, PDF p. 985)

180. c The error code comes from the separate circuit that runs the automatic system check by built-in test. (L3, PDF p. 985)

181. c The float of predetermined weight submerges in accordance with the density, so a deeper float means a different density. (L3, PDF p. 987)

182. b The system integrates a preselection of the desired volume together with a shutoff on reaching it or shortly before overflowing. (L3, PDF p. 987)

183. b Shutoff occurs on reaching the preselected value or shortly before overflowing, which depends on aircraft position. (L3, PDF p. 987)

184. a Cadensicon and compensator data are used to calculate the density of the fuel in the tanks. (L3, PDF p. 989)

185. a The computer needs both the volume and the fuel density before it can calculate the fuel mass. (L3, PDF p. 989)

186. c The computer can calculate a mean value from the remaining fuel and the replenished fuel. (L3, PDF p. 989)

187. a Fuel consistency differs between countries, so the remaining and replenished fuel can have different K to D values that the computer has to average. (L3, PDF p. 989)

Ultrasonic measurement, densitometers and testing

PDF pages 991–995 · 21 questions

LEVEL 1 — RECALL

188. In a liquid level metering system that works by timing an echo, which kind of signal do the tank units transmit ●

a. An ultrasonic signal
b. A radio frequency signal
c. A light signal

189. A tank of an ultrasonic system is opened for inspection, so how many tank units will be found in it? ●

a. A single unit
b. Numerous units
c. Exactly two units

190. A technician surveys an ultrasonic fuel tank installation, so how many densitometers will each tank have? ●

a. Two
b. One for every tank unit
c. One

191. Where are the water detectors fitted ●

a. At the low point in each fuel tank
b. In the refuelling gallery pipework
c. At the highest point of each fuel tank

192. On which kind of quantity measuring system is the GF-2 fuel tester used ●

a. Ultrasonic measuring systems
b. Float operated gauging systems
c. Capacitance measuring systems

LEVEL 2 — UNDERSTAND AND EXPLAIN

193. Where is the signal sent out by an ultrasonic tank unit reflected ●

a. At the surface of the liquid
b. At the bottom of the tank sump
c. At the wall of the stillwell tube

194. What picks up the echo in an ultrasonic quantity metering system ●

a. A separate receiver at the tank top
b. The tank unit that sent the signal
c. The compensator near the tank bottom

195. A computer has received the echo from an ultrasonic tank unit, so what does it use to find the liquid level? ●

a. The frequency shift of the returned echo
b. The running time of the signal
c. The strength of the returned echo

196. What does the fuel quantity processor unit measure in order to find the fuel height ●

a. The travel time of the pulse
b. The frequency of the returned pulse
c. The strength of the returned pulse

197. What does the sonic pulse travel through on its way to the fuel surface ●

a. The fuel in the stillwell
b. The wall of the tank unit
c. The air space above the fuel

198. Some tank units also carry calibration targets, so what does the processor unit get from the pulses that come back off them?

a. The depth of water in the tank sump
b. The density of the fuel in the tank
c. The velocity of sound in the fuel

199. Which kind of transducer does a densitometer use ●

a. A cardanically suspended float
b. A capacitive condenser
c. A vibrating cylinder

200. A densitometer output frequency is being read off, so what is that frequency proportional to? ●

a. The fuel height
b. The fuel density
c. The fuel temperature

201. How does a water detector show that water is lying in the bottom of a tank ●

a. Its pulse is absorbed so that no echo returns at all
b. Its pulse reflects off the boundary with the fuel
c. Its float sinks in the water and drives a transformer

202. A technician is told the water detectors work in the same way as the tank units, so what kind of devices are they? ●

a. Capacitive devices
b. Ultrasonic devices
c. Resistive devices

203. Which part of the test equipment must never come into contact with the airframe during a capacitance check

a. The earth strap of the tester
b. The connector of the tank unit
c. The casing of the tester

LEVEL 3 — APPLY AND REASON

204. Unlike an ultrasonic tank unit, which property of the fuel does the capacitance of a tank unit respond to ●

a. Its dielectric constant
b. Its density in the tank
c. Its temperature in flight

205. An ultrasonic system has to display a fuel weight, so which two measurements must the processor unit bring together? ●

a. The fuel density and the velocity of sound
b. The fuel height and the fuel temperature
c. The fuel height and the fuel density

206. A densitometer in one tank of an ultrasonic system has failed, so which calculation is lost first? ●

a. The fuel height in that tank
b. The fuel weight for that tank
c. The water level in that tank

207. A capacitance quantity system has to be checked on the ground with the tanks drained, so what does the tester let the technician do instead of wetting the probes? ●

a. Simulate the capacities
b. Simulate the fuel density
c. Measure the contact resistances

208. A tank unit reads correctly on the tester but the cockpit indication is still wrong, so which further tester function would be used on the wiring? ●

a. Simulating the probe capacities
b. Measuring contact resistances
c. Measuring the tank unit currents

Answers — Ultrasonic measurement, densitometers and testing

188. a The tank units transmit an ultrasonic signal. (L1, PDF p. 991)

189. b Each tank contains numerous tank units. (L1, PDF p. 991)

190. c Each fuel tank has one densitometer. (L1, PDF p. 993)

191. a Water detectors are at the low point in each fuel tank. (L1, PDF p. 993)

192. c The tester is used for capacitance measuring systems for fuel and oil. (L1, PDF p. 995)

193. a The signal is reflected at the surface of the liquid. (L2, PDF p. 991)

194. b The same tank unit that transmits also receives the echo. (L2, PDF p. 991)

195. b The computer calculates the liquid level from the running time of the signal. (L2, PDF p. 991)

196. a The processor unit measures the travel time of the pulse and uses it to find the fuel height. (L2, PDF p. 993)

197. a The pulse goes through the fuel in the stillwell to the fuel surface and back to the receiver. (L2, PDF p. 993)

198. c Pulses reflected from the targets are used to measure the velocity of sound in the fuel. (L2, PDF p. 993)

199. c Densitometers use a vibrating cylinder transducer. (L2, PDF p. 993)

200. b The frequency of the vibrations is proportional to the fuel density. (L2, PDF p. 993)

201. b If water is present the pulse reflects off the interface between the water and the fuel. (L2, PDF p. 993)

202. b The water detectors are ultrasonic devices like the tank units. (L2, PDF p. 993)

203. c The tester casing must never come into contact with the airframe. (L2, PDF p. 995)

204. a Capacitive measurement works from the differing dielectric constants, while the ultrasonic unit works from a reflected signal. (L3, PDF p. 991)

205. c Height comes from the tank units and density from the densitometer, and the FQPU calculates the fuel weight from them. (L3, PDF p. 993)

206. b Height still comes from the tank unit, but without the density from the densitometer the processor unit cannot work out that tank's fuel weight. (L3, PDF p. 993)

207. a The tester can simulate capacities as well as measure them. (L3, PDF p. 995)

208. b Besides measuring and simulating capacities the tester measures contact resistances, which is what a wiring check needs.
(L3, PDF p. 995)